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Changes in access to basic needs among people experiencing unsheltered homelessness in Los Angeles County during and after the COVID-19 pandemic: a large-scale repeated cross-sectional survey

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Abstract

Background People experiencing unsheltered homelessness (PEUH) face ongoing barriers to accessing basic resources, such as food, water, sanitation, and healthcare. While it is well-understood that the COVID-19 pandemic further exacerbated pre-pandemic conditions, less is known about how conditions shifted once the pandemic subsided.

Methods We employed a repeated cross-sectional, modified version of the HOUSED BEDS survey among a convenience sample of PEUH ($N = 991$) in the greater Los Angeles area between January 1, 2022 and December 31, 2024 to assess access to basic resources for survival by people experiencing unsheltered homelessness during and after the COVID-19 pandemic. Univariate and bivariate analyses were conducted.

Results Following the pandemic, several significant differences in access to basic resources were observed between the during- and post-pandemic samples: food and water security decreased (Food: 70.8% during vs. 61.2% post, $p = 0.002$; Drinking water: 77.2% vs. 67.6%, $p = 0.001$), as did having health insurance (83.9% vs. 75.8%, $p = 0.009$). However, self-reported incidences of open defecation were also observed to be lower in the population surveyed following the pandemic (90.4% during vs. 81.4% post-pandemic, $p < 0.001$). Relatedly, a correlated increase in reported bathroom access (20.6% vs. 26.0%, $p = 0.053$) and decrease in primary care access (18.6% vs. 14.8%, $p = 0.2$) were observed. Outreach services were provided to few respondents throughout (34.5%); however, rates of physical health outreach were found to be lower in the post-pandemic group (10.5% vs. 6.7%, $p = 0.032$) despite all other forms of outreach experiencing little change over this same time period.

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Conclusion These findings highlight the importance of continuing to support PEUH even after the pandemic has subsided. Implementing more long-term, institutionalized policy and infrastructure is essential for supporting this population during and after public health emergencies.

Keywords COVID-19 Pandemic, Unsheltered Homelessness, Basic Needs, Street Medicine

Background

The homelessness crisis in America continues to grow in urgency. The most recent counts estimate 771,480 people were experiencing homelessness nationwide in 2024; representing consecutive increases of 12.1% and 18.1% over the previous two years despite relative stability over the last decade [1]. As a result, understanding the needs of this growing population is a necessary step in creating sustainable, long-term infrastructure and policy to support them. Increasingly, studies are documenting the heterogeneity of the homeless population by distinguishing between "sheltered" and "unsheltered" homeless populations [2–5]. While "sheltered" homelessness refers to those engaged in services offered within the walls of homeless shelters, people experiencing unsheltered homelessness (PEUH) are defined as individuals who "live outside or in places not meant for human habitation, like a car, tent, or abandoned building" [6]. Compared to sheltered counterparts, PEUH consistently face significantly greater difficulty meeting basic needs such as food, water, sanitation, and access to healthcare [7, 8]; many or all of which are provided by homeless shelters. As a result, PEUH are over 25 times more likely than sheltered homeless populations to suffer from tri-morbidity, or co-occurrence of a physical health condition, mental health issue, and a substance use disorder [8]. Being among the most marginalized and vulnerable populations, PEUH's ability to meet basic survival needs serves as a critical barometer for the effectiveness of public health and social policies.

In Los Angeles (LA) County, the majority (70%) of people experiencing homelessness are living unsheltered, with the streets of the city of LA being home to the greatest number of PEUH in the nation [1, 9]. Between 2020 and 2021, the COVID-19 pandemic prompted an unprecedented volume of emergency orders and directives from local, state, and national leaders that specifically aimed at prioritizing public health and safety as well as broadly increasing the public's access to basic needs through social service expansions. These expansions included increased Supplemental Nutrition Assistance Program (SNAP) eligibility and benefits [10, 11] and increased access to free medical services, including vaccinations regardless of insurance status [12]. While these changes positively impacted PEUH, other measures for public safety, such as closures of non-essential businesses, public parks, and public spaces [13] negatively and disproportionately affected PEUH and their ability to access basic resources for survival. Together, these factors allow for a unique opportunity to simultaneously evaluate the effects of various types of policies on PEUH.

Despite policy-related efforts, it has been well-established that PEUH experienced disproportionate negative effects from the pandemic and even heightened pre-existing barriers to basic needs [14, 15] which have been reflected in this population's health outcomes on a large scale [16–19]. However, it is unclear how these conditions have changed, particularly following the withdrawal of temporary COVID-19 support measures. Despite a growing body of small-sample, qualitative research studies characterizing barriers to the basic needs of PEUH [14, 15, 20, 21], there is a lack of large-scale, population-based investigations focused on the COVID-19 pandemic period and the transition period thereafter. The purpose of this study was to evaluate access to basic survival resources (food, drinking water, sanitation, hygiene, homeless service outreach, health insurance, and primary care) by PEUH in LA County during and after the COVID-19 pandemic.

Methods

We conducted a repeated cross-sectional assessment survey over three years (January 1, 2022, to December 31, 2024) in LA County among a convenience sample of PEUH at the time of care initiation with USC Street Medicine. Data were collected prospectively on a continuous basis as part of routine clinical care and later analyzed retrospectively for research purposes. This study was reviewed and approved by the Institutional Review Board (IRB) of the University of Southern California (USC) and classified as exempt research (approval number UP-22-00587); not requiring written consent for the research evaluation. Research was conducted in accordance with the World Medical Association Declaration of Helsinki. At the time of care, participants were informed verbally by trained study staff about the purpose of the survey, the voluntary nature of participation, and how their information would be used. Additionally, participants provided written consent to receive medical care from USC and to allow deidentified health data (such as those reported here) to be used retrospectively for research, quality improvement, and publication. Patients also had the option to consent to medical care while declining the use of their data for research or quality improvement purposes. Patients were explicitly informed that refusal to participate in the survey or have their data used would not affect their access to care, the quality of care received, or any services provided.

Procedures and sample

Members of the USC Street Medicine team—including physicians, physician assistants, nurses, and community health workers—conducted convenience sampling by verbally administering the survey to all patients during their initial visit when establishing care with USC Street Medicine. Patient responses were self-reported and entered directly by team members into Qualtrics without the inclusion of personal identifiers. Eligible participants were adults aged 18 years or older who self-identified as experiencing unsheltered homelessness at the time of contact and agreed to engage in care with Street Medicine; no exclusion criteria were applied. It is worth noting that participants were not "recruited" in the traditional sense but were, rather, actively approached by the USC Street Medicine team in accordance with broader Street Medicine practices in order to minimize self-selection bias.

Although patients were followed longitudinally for healthcare purposes, the survey data presented here is derived from a repeated cross-sectional design, where each cross section represents a distinct individual, and were analyzed retrospectively. Follow-up assessments are conducted periodically as part of routine care when a patient's condition or living circumstances change, but these subsequent data are not systematically recorded or included in the research dataset.

Measures

The in-person baseline health assessment utilized a modified version of the *HOUSED BEDS* [22] clinical tool, designed specifically to collect comprehensive health and social histories from individuals experiencing unsheltered homelessness. The original HOUSED BEDS instrument was adapted by its creators in collaboration with clinical members of the USC Street Medicine team and pilot tested by three team members with 54 patients between November and December 2021. The pilot revealed no need for further revisions, and the instrument was implemented into usual care beginning in 2022. The acronym *HOUSED BEDS* represents the domains of *Homeless history, Outreach, Utilization, Salary, Eat, Drink, Bathroom, Encampment, Daily routine*, and *Substance use.* A detailed description of the tool and its components is publicly available. The modified version used in this study is modified to focus on access to basic needs, defined as food, water, sanitation, and basic healthcare. Accordingly, our modified *HOUSED BEDS* survey included 14 items (plus two with skip logic) directly addressing outreach, healthcare utilization, eating, drinking, and bathroom access. The assessment also included three self-reported demographic questions (age, gender, and race/ethnicity). The complete survey is provided in Appendix A.

Analysis

Univariate statistics (frequencies, percentages, means, and standard deviations) were used to describe all survey variables for the overall sample and for the COVID-19 pandemic and post-pandemic periods. The study employed a repeated cross-sectional design where surveys were administered over the course of three years; each of the $n = 991$ surveys represents a unique participant, with no repeated measures on the same individuals. A date variable for "During" vs. "Post-pandemic" was created based on whether the survey was administered on or after May 5, 2023, the official end date of pandemic as designated by the WHO [23]. This date is used as a standardized and externally defined cutoff to distinguish between the periods, providing a consistent and reproducible reference that has been widely used in public health research.

Bivariate analyses were conducted, including one-way ANOVA for continuous variables and chi-square tests for categorical variables. To determine the extent to which a shift in demographic factors influenced changes in patterns of needs after the pandemic, we performed logistic regression models for the effect of pandemic on each outcome, adjusting for demographic characteristics. We used $\alpha=0.05$ for all statistical tests. Analyses were conducted in R statistical software (v4.4.2). Data was labeled as "missing" if a response was not provided for the particular measure(s); deletion, imputation, or estimation were not conducted. All analyses were performed retrospectively by a member of the study team with expertise in biostatistics, using R statistical software.

Results

Demographic characteristics

A total of 991 patients/respondents experiencing unsheltered homelessness were surveyed across multiple regions of LA County at the time of care initiation with street medicine. Most were seen/surveyed by the street medicine team covering LAs' Eastside (34.1%), followed by South LA (18.2%), Hollywood (14.9%) and Council District 1 (14.8%). Overall, the population was predominantly male (65.2%) or female (33.9%) with a few identifying as transgender (0.8%, $n = 8$) or nonbinary (0.1%, $n = 1$). Respondents' racial and ethnic background was primarily Hispanic (42.7%), followed by White (27.6%), Black (23.0%), and Other/Multiple ethnicity (4.8%). The mean age was 45 years (SD = 13), with participants ranging in age from 18 to 80 years old. Demographic information is summarized in Table 1.

Table 1 Demographic characteristics

Characteristic	Freq (%)
	***N*=991**
Patient Group	
COLA-Eastside	338 (34.1%)
COLA-South LA	180 (18.2%)
COLA-Hollywood	148 (14.9%)
COLA-CD1	147 (14.8%)
SPA 3	122 (12.3%)
Mid City	56 (5.65%)
Gender Identity	
Male	638 (65.2%)
Female	332 (33.9%)
Transgender	8 (0.82%)
Nonbinary/nonconforming	1 (0.10%)
Race/Ethnicity	
Asian or Pacific Islander	16 (1.80%)
Black or African American	205 (23.0%)
Hispanic	380 (42.7%)
Multiple ethnicity or Other	43 (4.83%)
White or Caucasian	246 (27.6%)
Age (Years)	
Median (Q1,Q3)	43 (34, 56)
Mean (SD)	45 (13)
Min, Max	18, 80

Unmet needs and survival indicators

Compared to the participants surveyed during the pandemic (70.8%), only 61.2% of participants surveyed following the pandemic reported eating daily after the pandemic ($p = 0.002$). This was defined as eating greater than or equal to one meal in a 24-hour period, where a meal was defined as a structured eating occasion typically corresponding to breakfast, lunch, or dinner. Similarly, access to drinking water, as defined by individual participants, was significantly lower in the post-pandemic group, with 77.2% in the during-pandemic group and 67.6% in the post-pandemic group ($p = 0.001$). The means of obtaining drinking water shifted to a greater reliance on shared infrastructure. For example, individuals were less likely to buy bottled water post-pandemic (39.1% during vs. 30.7% after) and more likely to utilize public water sources (27.7% during vs. 33.8% after; $p = 0.016$). In contrast, the proportion of individuals purchasing food increased from 46.3% during the pandemic to 52.7% post-pandemic, while those relying on donated food declined from 37.5% to 30.9%, and those retrieving food from the trash decreased from 5.2% to 2.7% ($p = 0.067$). After adjusting for demographic characteristics, the odds of participants eating daily and having access to drinking water were still lower after the pandemic vs. during (OR = 0.73, 95% CI = [0.53, 0.99] for eating daily; OR = 0.65, 95% CI = [0.46, 0.91] for drinking water).

Sanitation indicators showed significant improvement following the pandemic. The percentage of participants surveyed during the pandemic who reported having to resort to open defecation was observed to be significantly greater than the percentage of participants surveyed following the pandemic (90.4% during vs. 81.4% after; $p < 0.001$). Relatedly, the proportion of individuals reporting having access to a bathroom rose from 20.6% during the pandemic to 26% post-pandemic ($p = 0.053$). Adjusting for demographic characteristics, individuals were still less likely to resort to open defecation after the pandemic vs. during (OR = 0.46, 95% CI = [0.28, 0.72]). See Table 2.

Medical access

Our modified HOUSED-BEDS survey included three items regarding health insurance status, primary care access, and primary care utilization. Unmet healthcare needs remained substantial, and in some cases, worsened, following the pandemic. Health insurance coverage declined significantly, from 83.9% during to 75.8% after the pandemic ($p = 0.009$). A strong association was observed between insurance and primary care provider (PCP) access, as only 1.2% of individuals (2/164) without insurance reported having a PCP, compared to 22.7% (128/564) of those with insurance ($p < 0.001$). A correlated decline was found in the proportion of participants with a PCP from 18.6% during the pandemic to 14.8% post-pandemic ($p = 0.2$). The effect of the pandemic on health insurance coverage persisted after adjusting for demographic characteristics (OR = 0.61, 95% CI = [0.40, 0.90] for during- vs. post-pandemic). See Table 3.

Outreach

Respondents were asked about receipt of physical health, mental health, housing, or other type of outreach in the past two weeks. Results showed that compared to the population surveyed during the pandemic, outreach trends were notably different in the population surveyed after the pandemic. Overall, about a third (34.5%) of the study sample reported receiving any type of outreach in the past two weeks (data not shown). Among all surveyed participants, housing outreach was the most common (21.5%), followed by physical health (8.27%), and mental health (3.23%). However, the type of outreach offered during and after the pandemic changed significantly; physical health outreach declined (10.5% during the pandemic to 6.70% post-pandemic, $p = 0.032$), while housing related outreach increased from 17.8% during- to 24.1% post-pandemic ($p = 0.019$). The decline in outreach for physical health was still statistically significant after adjusting for demographic characteristics (OR = 0.43, 95% CI = [0.23, 0.80]). Mental health and 'other' outreach remained relatively stable, showing modest increases over the study period. See Table 4.

Table 2 Access to basic needs among adult PEUH in LA County, by survey period (During vs. Post–COVID-19), with odds ratios for the effect of thepandemic, adjusting for age, gender, and race/ethnicity

Characteristic	Frequency (%)				OR (95% CI)^2
	Overall	During Pandemic	Post-Pandemic	*p*-value^1	
	N **= 991^1**	***N*** **= 409^1**	***N*** **= 582^1**		
Access to Bathroom					
	229 (23.8%)	81 (20.6%)	148 (26.0%)	0.053	1.20 (0.86, 1.68)
Does Not Defecate Outside					
	136 (14.9%)	36 (9.63%)	100 (18.6%)	< 0.001***	2.19 (1.40, 3.53)***
Access to Shower					
	380 (49.4%)	121 (46.9%)	259 (50.7%)	0.3	1.17 (0.86, 1.59)
Eats Daily					
	616 (65.2%)	279 (70.8%)	337 (61.2%)	0.002**	0.74 (0.54, 1.00)
Has Drinking Water					
	674 (71.5%)	301 (77.2%)	373 (67.6%)	0.001**	0.66 (0.48, 0.92)*
Does Not Use Hydrant					
	460 (69.1%)	128 (76.2%)	332 (66.7%)	0.021*	0.62 (0.40, 0.93)*
Does Not Worry Water					
	386 (60.0%)	107 (64.1%)	279 (58.6%)	0.2	0.82 (0.56, 1.19)

*$p<0.05$; **$p<0.01$; ***$p<0.001$

^1 Pearson's Chi-square test

^2 Odds ratio for post- vs. during-pandemic, adjusting for demographics

Table 3 Health Insurance and primary care access among adult PEUH in LA County, by survey period (During vs. Post–COVID-19), with odds ratios for the effect of the pandemic, adjusting for age, gender, and race/ethnicity

Characteristic	Frequency (%)				OR (95% CI)^2
	Overall	During Pandemic	Post-Pandemic	*p*-value^1	
	N **= 991**	***N*** **= 409**	***N*** **= 582**		
Has PCP					
	135 (16.1%)	55 (18.6%)	80 (14.8%)	0.2	0.85 (0.57, 1.27)
Has Health Insurance					
	603 (78.6%)	224 (83.9%)	379 (75.8%)	0.009**	0.57 (0.38, 0.85)**

*$p<0.05$; **$p<0.01$; ***$p<0.001$

^1 Pearson's Chi-square test

^2 Odds ratio for post- vs. during-pandemic, adjusting for demographics

Table 4 Outreach among adult PEUH in LA County, by survey period (During vs. Post–COVID-19), with odds ratios for the effect of the pandemic, adjusting for age, gender, and race/ethnicity

Characteristic	Frequency (%)				OR (95% CI)^2
	Overall	During Pandemic	Post-Pandemic	*p*-value^1	
	N **= 991**	***N*** **= 409**	***N*** **= 582**		
Outreach: Physical Health					
	82 (8.27%)	43 (10.5%)	39 (6.70%)	0.032*	0.40 (0.22, 0.73)*
Outreach: Mental Health					
	32 (3.23%)	11 (2.69%)	21 (3.61%)	0.4	1.03 (0.46, 2.42)
Outreach: Housing					
	213 (21.5%)	73 (17.8%)	140 (24.1%)	0.019*	1.61 (0.95, 2.73)
Outreach: Other					
	52 (5.25%)	19 (4.65%)	33 (5.67%)	0.5	1.12 (0.58, 2.26)

*$p<0.05$; **$p<0.01$; ***$p<0.001$

^1 Pearson's Chi-square test

^2 Odds ratio for post- vs. during-pandemic, adjusting for demographics

Discussion

This study uniquely focused on the unsheltered homeless population in LA County to compare access to basic survival resources within a convenience sample that was taken during and after the COVID-19 pandemic. Findings provide novel evidence regarding availability of food, drinking water, toilet, outreach, health insurance, and primary care for PEUH during an unprecedented period of enactment and withdrawal of numerous pandemic welfare policies. Our findings suggest a wide variety of general public policies, ranging from nationwide continuous Medicaid coverage requirements [24] to shutting off local public water sources [13], disproportionately affected PEUH, whether for better or for worse. Despite being affected more directly than other populations, PEUH are rarely specifically accommodated or even considered when these policies are drafted [25]. Our findings may provide suggestive evidence to inform policies that accommodate the basic needs of the particularly marginalized PEUH population that are underrepresented in homelessness research.

Our study revealed greater food and water insecurity among the post-pandemic group when compared to the during pandemic group. Food insecurity refers to limited access to sufficient nutritious food, while water insecurity refers to unreliable access to clean, safe water. A possible explanation for these findings was the substantial changes to Supplemental Nutrition Assistance Program (SNAP), a federal "food stamp" program that provides financial assistance to low-income individuals and families, at both the federal and state levels during this time. While pandemic-era increases in monthly SNAP benefits from \$194 to \$291 [26] have remained in place at the time of data collection, other temporary pandemic measures, such as the waiving of certification/recertification interviews and simplified income reporting requirements [10, 11] have been discontinued in March 2023. This may have resulted in recipients abruptly losing this benefit. These expansions may be supported by our data suggesting increasing practices in directly purchasing food and decreases in utilization of donations, trash cans, and food banks. They may be additionally supported by the finding that purchased bottled water was the largest source of drinking water for PEUH by a large margin during the pandemic (39.1%).

This observed difference in the proportion of participants obtaining drinking water from purchased and donated bottled water between the during- and post-pandemic groups could be specifically related to the significant increase in the utilization of public water sources which were reopened after the pandemic. Previous research has suggested that public spaces have become de facto basic needs infrastructure despite not being designed for such purposes [27], often resulting in the failure to acknowledge the reliance of PEUH on the functionality of these public spaces for survival. These findings are then perhaps unsurprising due to county-wide water shut-offs, largely impacting public drinking fountains [13], during the pandemic which were revived following the pandemic. Similarly, public restroom closures during the pandemic serve as one possible explanation for significant decreases in open defecation, a commonly recognized measure of sanitation access [27]. It is worth noting that, despite the post-pandemic group reporting lower rates of open defecation, rates of open defecation remain high in both the during- and post-pandemic groups (90.4% and 81.4%, respectively). This is consistent with findings from Capone et al., 2020, which found that the degree of sanitation deprivation far exceeded official estimates from the WHO/UNICEF Joint Monitoring Program for Water Supply, Sanitation and Hygiene [28, 29], where PEUH were excluded from the estimates. There are a number of possible alternative explanations for the increase in sanitation access, such as encampment management policies, availability of portable sanitation infrastructure, or relocation of individuals following the pandemic; however, only restroom closures are addressed here due to its thorough legal documentation and direct relevance to sanitation access.

Taken together, significant changes in access across some of the most fundamental basic needs following the drastic changes in public policy during the COVID-19 pandemic may suggest the disproportionate impact of changes in public resources on PEUH [30]. However, despite this disproportionate impact, it is evident that PEUH are excluded from public discourse as seen by the complete and immediate cutting off of some of this population's major sources of basic needs in the face of emergency. Although these emergency directives may not be malicious, they demonstrate an ignorance regarding basic needs access among the PEUH population and the necessity of large-scale approaches in characterizing the needs of this population. More rigorous approaches are needed to determine the extent of the various public policies on the PEUH population.

Our study reveals low healthcare utilization among PEUH in LA County and suggests further significant decreases in insurance coverage and correlated decreases in PCP utilization following the pandemic. However, universally, it is found that healthcare for PEH costs the public significantly more than housed individuals with one study finding that, even after adjustment for age, gender, and resource intensity weight, unhoused patients cost over 1.5 times more per hospital admission on average [31]. Despite an already alarming discrepancy, this average value is likely significantly blunted as studies increasingly find PEUH to be outliers in the general PEH population [32], largely due to the significantly increased prevalence of tri-morbidities and other compounding health complications in this sub-population [2, 3, 33].

A major contributing factor to the compounding nature of PEUH health complications is the lack of utilization of preventative care such as regular appointments with PCPs [7, 20]. Our study observed significantly lower rates of insurance coverage in the post-pandemic group, consistent withdrawal of pandemic-era Medicaid expansions. These expansions primarily consisted of widened eligibility criteria, such as nationwide continuous enrollment protections where states were required to keep people enrolled in Medicaid regardless of eligibility changes throughout the pandemic (which occurred on March 31, 2023), in order to ensure uninterrupted access to healthcare during the public health emergency. However, among PEUH, our findings reveal that, although there is a robust one-way correlation between insurance coverage and PCP utilization, PCP utilization remained low among participants surveyed both during and after the pandemic despite the majority of participants possessing medical insurance in both the during- and post-pandemic groups. This may suggest that lack of preventative care utilization is not entirely attributable to cost or ability to pay. The most widely-reported barrier is competing priorities for survival, such as meeting a daily basic need for food, water, and toilet [7, 20]. One study finds that PEH experiencing frequent difficulty meeting these basic needs were about one third as likely as those with infrequent difficulty to utilize preventative care and almost twice as likely to have gone without needed medical care [21]. Accordingly, our study characterizes in greater detail the specific impact of public policy shifts on the ability of PEUH to meet basic needs which is a crucial prerequisite to effective and urgently-needed PCP utilization. This basic needs-centered explanation is additionally supported by the lower proportion of PCP utilization in the post-pandemic group which may be due in part to the massive expansion and subsequent contraction of the street-based, PEUH-centered COVID-19 workforce, which provided medical care in the streets, regardless of PEUH's access to basic needs [12, 34].

Our study found that physical health outreach, specifically targeting PEUH, was dramatically lower in the post-pandemic group while all other forms of outreach were higher when compared with the during-pandemic group (some statistically insignificant). Of the previously mentioned policies thus far (SNAP, Medicaid, etc.), all involve sweeping changes affecting the general public with none specifically considering the needs of PEUH. However, previous studies have shown that substantial quantities of PEUH do not benefit these generalized programs due to exclusionary practices that are incompatible with unsheltered status [35–37]. For example, one large-scale study in LA among food pantry users, found that only 5.7% of PEH received food stamps, with the number likely being even lower for PEUH, despite often being eligible for the maximum amount [38]. During the pandemic, state and local governments were forced to acknowledge these structural inefficiencies due to the growing view of PEUH living conditions—lack of shelter for sheltering-in-place, lack of hygiene and personal protective equipment (PPE) resources, prevalence of comorbidities, low vaccination status, etc.—as a threat to public health during the COVID-19 pandemic [39–41], allowing for a rare opportunity to assess the effectiveness of large-scale, street-based health outreach efforts that specifically targeted PEUH. In LA County, the LA County Department of Health Services and LA County Department of Public Health implemented an unprecedented joint mobile vaccination program, mobilizing large interdisciplinary healthcare teams to facilitate 2658 vaccine clinics for PEH, 1626 (61.2%) of which specifically targeted PEUH, in LA County in 2021 alone [12]. The effects of this single year of unprecedented physical health outreach is reflected in our data as seen by the significantly lower proportions of physical health outreach in the post-pandemic group which is consistent with the termination of these initiatives (33.1% to 19.7%, $p = 0.003$). Despite the transience of these pandemic initiatives due to the nature of emergency funding, these initiatives demonstrate that the limited access to basic needs among PEUH can be alleviated by initiatives that specifically consider barriers that are unique to this population. As of now, PEUH continue to lack lasting infrastructure for the delivery of healthcare at a scale that is commensurate with the size of the PEUH population.

Street Medicine is emerging as a potential model for addressing this infrastructure gap in a PEUH-specific manner. Street Medicine is defined as "health and social services developed specifically to address the unique needs and circumstances of the unsheltered homeless delivered directly to them in their own environment" [42] and has been repeatedly shown to be effective in successfully housing PEUH, reducing emergency room costs, and fostering trust with PEUH [43–45]. While our study demonstrates the need for the recognition of PEUH in public policy, we recognize that this may be difficult, especially in emergency situations, such as the pandemic, where immediate large-scale, executive actions must be made for public safety. However, through the interdisciplinary nature that is inherent to Street Medicine, Street Medicine is able to serve as a consistent PEUH-specific service that is able to advocate for PEUH in any given social context by comprehensively providing access to basic needs as well as downstream needs such as housing and employment. For example, by providing care to PEUH in their own environments, Street Medicine is able to overcome non-financial barriers to primary care providers that PEUH consistently face [46]. Regarding food and water, Street Medicine is able to provide case management services to help navigate the complex SNAP services [46]. Though the few studies in the literature that describe Street Medicine suggest it to be a promising model of healthcare and social service delivery, more rigorous analyses are needed to further explore this model's effectiveness and scalability in advocating for the interests of PEUH which have been historically excluded from public policy.

Limitations

This large-scale repeated cross-sectional survey uniquely targeted PEUH over three years during and after the COVID-19 pandemic but it is not without limitations. First, this study relied on self-reported data from PEUH initiating street medicine care with a single organization covering a particular geographic service area in LA County. Because some PEUH subpopulations may be less accessible or willing to engage with USC Street Medicine, findings may not be representative of the larger unsheltered homeless population in LA, or generalizable to other areas. However, we found similar demographic characteristics among our participants compared to those assessed in the latest (2024) annual Point-in-Time Count (PIT Count) [1]. Data were cross-sectional and lacked longitudinal follow-up; precluding our ability to explore individual and cohort changes over time. Additionally, our study lacks data from prior to the pandemic which prevents important analyses of pandemic policy impact. Finally, the use of a single time point (May 5, 2023, the WHO-designated end of the Covid-19 pandemic) to create a binary distinction between the during- and post-pandemic groups may not reflect the timing of differing federal and local policy changes that affected PEUH, which occurred gradually and varied by jurisdictions within Los Angeles County. As a result, comparisons between periods should be interpreted accordingly. Despite these limitations, this study is among the first to target PEUH, recruit a large sample of PEUH, and elicit their experience during a multi-year study period.

Conclusions

Our study suggested significant changes in access to basic survival resources for PEUH from during, to after, the COVID-19 pandemic: a turbulent time which holds important implications for personal health, wellbeing, and outcomes of PEUH and broader public health.

These changes may be related to significant shifts in public policy during this time period, including withdrawal of emergency welfare benefits and initiatives and the re-opening of public spaces, many of which are known to have disproportionately affected PEUH. The elucidation of the effects of these policies on PEUH in additional research is important to allow for a more informed approach in developing future policies that better recognize and serve PEUH as part of community and public health.

Abbreviations

EBT	Electronic Benefits Transfer
LA	Los Angeles
PEH	People Experiencing Homelessness
PEUH	People Experiencing Unsheltered Homelessness
PCP	Primary Care Provider
SNAP	Supplemental Nutrition Assistance Program

Supplementary Information

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Supplementary Material 1

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Authors' contributions

SY conducted data analysis and interpretation as well as drafting, editing, and revising the manuscript. AI and CA contributed select sections of the initial manuscript draft. TP performed all statistical analyses. ACK and CF provided senior oversight and revised the manuscript. All authors reviewed and approved the final version.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was reviewed and approved by the institutional review board (IRB) of the University of Southern California as a secondary data analysis meeting the classification for exempt research (UP-22-00587). Participants provided signed (i.e. written) informed consent to engage in medical care with USC Street Medicine (an entity of Keck Medicine of USC) and have deidentified health data (such as those reported here) analyzed for secondary research and quality improvement, and publication. Data analyzed in this study were originally collected as part of usual clinical care.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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